



INTERNSHIP REPORT

FIBER OPTIC COMMUNICATIONS, FTTH NETWORK ARCHITECTURE AND HOME APPLIANCES MAINTENANCE

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CERTIFICATE

This is to certify that the internship report entitled “Fiber Optic Communications, FTTH Network Architecture and Electrical Home Appliances Maintenance” is a bona fide record of the industrial training carried out by **HAMNA FATHIMA MP** under my academic supervision and guidance, in corporate collaboration with Bfone Fibernet, Edavannappara, in partial fulfillment of the requirements for the award of the Degree of Bachelor of Science

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Internship Activity Schedule

The day-wise internship activities and topics covered during the training period are summarized in the table below:

Day	Date	Topics / Activities Covered
Day 1	21.05.2026	Optical Fiber Theory Class (Introduction to OFC)
Day 2	22.05.2026	Splicing Practical Section
Day 3	23.05.2026	Repairing of the Ceiling Fan
Day 4	25.05.2026	• Theory Class of Fiber To The Home (FTTH) • Repairing of Mixer Grinder • Field Visit of Splicing
Day 5	26.05.2026	Field Visit at Manjeri Telephone Exchange
Day 6	08.06.2026	Power Calculation Theory Class
Day 7	09.06.2026	• Repairing of Iron Box • Repairing of Wall Fan • Repairing of Induction Cooker

Table 1: Day-wise Internship Activity Log

Part I

Fiber Optic Communications and FTTH Network Architecture

Part Introduction: Fiber Optic Infrastructure

Today's telecommunication sector is heavily reliant on high-speed data transmission networks built on principles of optical physics. Moving beyond legacy copper-wire networks, optical fibers guide light signals to transport large amounts of data over vast distances with minimal signal degradation. This part of the internship report establishes a detailed technical study of modern Fiber-Optic Communication (OFC) and Fiber-to-the-Home (FTTH) network architectures, directly informed by industrial training with Bfone Fibernet. The core purpose here is to outline the entire journey of the light signal, from the hardware source down to the home ONT. The chapters herein analyze the physical principles of Total Internal Reflection (TIR) that guide light within glass matrices, the structural mechanics of modern multi-core cables, and the quantitative engineering required for link budgeting and structural calculations. Furthermore, this section details practical operational protocols, including high-precision fusion splicing, color-coding standards, network diagnostics, and troubleshooting methodologies implemented in active field environments. The module wraps up with observations made during the field visit to the local BSNL Exchange.

Chapter 1

Introduction to Optical Fiber Communication

1.1 Overview

Fiber-optic communication (OFC) involves transmitting data from a source to a destination by sending light pulses through glass or plastic fibers. In this system, light acts as an electromagnetic carrier wave modulated to carry digital and analog data.

1.2 The Basic Fiber Optic Link

The basic structure of a fiber-optic communication link includes several essential elements:

- Encoder/Source: Converts data into electrical signals.
- Transmitter (E/O): Converts electrical signals into optical signals using LEDs or LASERs.
- Optical Fiber: The medium that guides the light signal.
- Receiver (O/E): Uses a photo-detector (PIN or APD) to convert light back into electrical signals.
- Decoder: Recovers the original data from the signal.

1.3 Advantages of Fiber Optics

- Non-conductive: No risk of sparks or electrical hazards.
- Electromagnetic Immunity: Immune to EMI and lightning.
- Large Bandwidth: Can carry massive amounts of data.
- Low Loss: Signal attenuation is significantly lower than copper cables.
- Small and Lightweight: Easier to install and transport.

Chapter 2

Principle and Cable Structure

2.1 Principle of Operation: Total Internal Reflection

The propagation of light along the fiber core is governed by the physical principle of Total Internal Reflection (TIR). The occurrence of TIR requires two strict conditions:

- The refractive index of the core (n_1) must be greater than the cladding (n_2).
- The angle of incidence must be greater than the critical angle (θ_c).

The mathematical formula for the critical angle is:

$$\theta_c = \sin^{-1} \left(\frac{n_2}{n_1} \right) \quad (2.1)$$

2.2 Cable Anatomy

A typical optical fiber cable is constructed with the following layers from the inside out:

- Core: The innermost glass or plastic part where light travels.
- Cladding: Layer with a lower refractive index that reflects light back into the core.
- Buffer: Protects the fiber from moisture.
- Strength Member: Usually Aramid yarn (Kevlar) to prevent cable breakage.
- Outer Jacket: Protects against environmental damage.

2.3 Cable Types

- Tight-Buffer Cable: Used mainly for indoor applications and patch cords.
- Loose-Tube Cable: Used for outdoor environments (ducts, direct burial).
- Ribbon Cable: Dozens of fibers packed together for high-density applications.

Chapter 3

Link Budgeting, Splicing, and Mathematical Calculations

3.1 OFC Splicing

Fiber splicing refers to creating permanent joins between two fiber cores. The primary techniques include:

- Fusion Splicing: Using an electric arc to melt and fuse fibers.
- Mechanical Splicing: Aligning fibers using a mechanical fixture.

3.2 Logarithmic Conversion Mechanics

The logarithmic decibel-milliwatt (dBm) scale is standard in fiber optics to simplify power loss calculations across links:

$$P_{\text{dBm}} = 10 \log_{10} \left(\frac{P_{\text{mW}}}{1 \text{ mW}} \right) \quad (3.1)$$

3.3 Splitter Loss Calculations

The ideal insertion loss of a passive optical splitter depends directly on the split ratio or output ports (N):

$$\text{Loss}_{\text{Ideal}} = 10 \log_{10}(N) \quad (3.2)$$

3.4 Standard Loss Values for Link Performance

The following standard loss values are utilized by technicians to calculate link budgets:

Network Element Component	Standard Loss Value
Fiber Path Loss @ 1310 nm / 1490 nm	0.40 dB per km
Fiber Path Loss @ 1550 nm	0.25 dB per km
Fusion Splice Junction	0.10 dB per splice
SC/UPC Physical Connector Pair	0.35 dB per connector
SC/APC Physical Connector Pair	0.25 dB per connector
1 × 2 Splitter Module	3.0 dB average insertion loss
1 × 16 Splitter Module	13.0 dB average insertion loss
1 × 32 Splitter Module	16.0 dB average insertion loss

Table 3.1: Standard Loss Parameters Used in FTTH Networks

3.5 Mathematical Verification: Power Link Budget Case Study

$$P_r = P_t - \text{Loss}_{\text{total}} \quad (3.3)$$

3.5.1 Case Study Parameter Blueprint

Let us examine a 7 km optical fiber link operated by Bfone Fibernet. This specific network span contains 4 fusion splices, 2 SC/UPC connectors, and two splitters in series: a 1 × 16 and a 1 × 8 splitter. The loss budget details for this specific fiber span are calculated below:

- Fiber Line Attenuation Loss: $7 \text{ km} \times 0.40 \text{ dB/km} = 2.8 \text{ dB}$
- First Stage Insertion Loss (1 × 16 Splitter) = 13.0 dB
- Second Stage Insertion Loss (1 × 8 Splitter) = 10.0 dB
- Total Mechanical Splice Losses: $4 \times 0.10 \text{ dB} = 0.4 \text{ dB}$
- Total Dynamic Connector Losses: $2 \times 0.35 \text{ dB} = 0.7 \text{ dB}$

The total insertion loss is calculated by adding the individual attenuation values:

$$\begin{aligned} \text{Total Attenuation Loss} &= 2.8 + 13.0 + 10.0 + 0.4 + 0.7 \\ &= 26.9 \text{ dB} \end{aligned}$$

3.5.2 Link Status Analysis

For reliable performance, typical ONT optical receivers must receive light power in the range of -10 dBm to -28 dBm: Minimum Receiver Threshold $\geq -28 \text{ dBm}$

$$\text{Receiver Sensitivity Threshold} \leq -28 \text{ dBm} \quad (3.4)$$

Since the total attenuation is 26.9 dB, assuming an OLT transmitter power of 0 dBm, the received signal power is -26.9 dBm. This is above the receiver threshold of -28 dBm, confirming that the fiber link is viable and will operate reliably.

Chapter 4

Optical Fiber Splicing: Detailed Analysis

4.1 What is Splicing?

Fiber splicing is defined as the technique of permanently connecting two optical fiber ends to form a seamless light path. While connectors are built for quick disconnects, splices are permanent joints optimized for stability. The primary goal is to minimize optical loss and prevent back-reflection.

4.2 Types of Splicing

The two principal splicing techniques are:

4.2.1 Fusion Splicing

Fusion Splicing is the standard technique in telecom networks. It uses a controlled electric arc to melt and fuse the glass fiber ends. Attenuation: Usually between 0.01 dB and 0.05 dB. Usage: Backbone networks, long-haul links, and FTTH.

4.2.2 Mechanical Splicing

Mechanical splicing aligns the two fiber ends inside a precision sleeve with index-matching gel to bridge the air gap. Attenuation: Usually ranges from 0.2 dB to 0.5 dB. Usage: Emergency repairs, temporary setups, and environments where fusion equipment is not available.

4.3 Importance and Significance

- **Link Extension:** Connecting multiple segments to build long-distance networks.
- **Signal Integrity:** Maintaining near-perfect light transmission with minimal attenuation.
- **Back-reflection Control:** Minimizing return loss, which protects laser sources.
- **Durability:** Providing maintenance-free connections that withstand environmental stress.

4.4 Step-by-Step Fusion Splicing Process

1. Stripping: The 250 μm outer coating is removed using specialized fiber strippers without scratching the glass.
2. Cleaning: The bare glass fiber is thoroughly cleaned using 99% Isopropyl Alcohol (IPA) and lint-free wipes to remove all contaminants.
3. Cleaving: The fiber is cut using a precision diamond-blade cleaver to ensure a perfect 90-degree end face (angle $\pm 0.5^\circ$).
4. Fusion: The fiber ends are aligned using the Profile Alignment System (PAS) and fused using an electric arc at 2000°C.
5. Protection: A heat-shrink splice sleeve is applied and heated to protect the joint from physical damage and moisture.

4.5 Advantages

- Lowest possible insertion loss.
- Permanent and highly reliable joints.
- Resistant to vibration and temperature fluctuations.
- Space-efficient for high-density fiber enclosures.

4.6 Future Scope

- Automation: Developing fully autonomous AI-driven splicing robots for sub-sea and hazardous environments.
- Multi-core Fiber Splicing: Moving beyond single-core to support high-density multi-core fibers to increase bandwidth capacity.
- Enhanced Precision: Integrating advanced nanotechnology for near-zero loss splicing in extreme environments.

4.7 Conclusion

In summary, splicing is fundamental to building reliable optical infrastructure. Creating permanent, low-loss connections ensures that gigabit data speeds are maintained over long spans. Fusion splicing remains the gold standard, while mechanical connectors provide critical utility for quick repairs.

Chapter 5

Introduction to FTTH & Passive Optical Networks (PON)

5.1 Overview of FTTH

Fiber-to-the-Home (FTTH) is the modern standard for high-speed internet delivery, replacing legacy copper telephone and cable lines completely with fiber lines run directly to consumer homes. The dominant technology in FTTH is PON, or Passive Optical Network. It is 'passive' because there are no powered switches in the field; light is divided using glass splitters. This design reduces power consumption, cuts down on hardware failures, and lowers operating costs. Through this point-to-multipoint topology, a single fiber feeds multiple households.

5.2 The FTTH Network Architecture Layers

The physical deployment of FTTH networks consists of three main segments: A. The Feeder Segment: Starts at the service provider's Central Office (CO) and runs to a local neighborhood hub known as the Fiber Distribution Hub (FDH) or Primary Cross-Connect box. This segment uses high-core-count cables (often containing 48 to 144+ fibers) designed to carry aggregated data over long distances (up to 20 kilometers) without splitting.

B. The Distribution Segment: Travels from the FDH to smaller street-level ODP (Optical Distribution Point) or FAT (Fiber Access Terminal) boxes. At the FDH, the feeder fibers are connected to splitters, and the split fibers branch out into the neighborhood using medium-core cables (12 to 24 fibers) along utility poles or underground conduits.

C. The Drop Segment: The final link that bridges the ODP/FAT on the street directly into the customer's living room or office. This utilizes a Drop Cable, which is a highly specialized, lightweight, ruggedized fiber cable (usually containing only 1 or 2 fibers) engineered to survive tight bends and physical pulling.

5.3 Core Hardware Components

The architecture is built on a close cooperation between powered active components and non-powered passive components:

5.3.1 Active Components (Power Required)

OLT (Optical Line Terminal): Installed at the Central Office (CO). It acts as the central 'brain' of the network, converting digital IP data from the backbone into laser pulses and scheduling traffic for thousands of subscribers.

ONT / ONU (Optical Network Terminal / Unit): Installed inside the subscriber's home. It acts as the translator, receiving optical light signals and converting them back into standard electrical signals (Ethernet/Wi-Fi) for consumer devices.

5.3.2 Passive Components (No Power Required)

Optical Splitters: The heart of PON. A passive glass prism component that takes a single incoming beam of light and splits it into multiple identical streams to serve multiple users. Standard split ratios are 1:8, 1:16, 1:32, or 1:64.

ODF / FDH Hubs: Environmental enclosures or cabinets that house splices and splitters, protecting connections from water, dust, and temperature changes.

Chapter 6

FTTH Deployment, Coding and Testing

6.1 Fiber Color Coding Standard (TIA-598-C)

To identify individual fibers in high-core-count cables, the industry uses a standardized 12-color coding sequence (TIA-598-C). For larger cables, the color sequence repeats using a black stripe or tracer.

No.	Fiber Color	Symbol	No.	Fiber Color	Symbol
1	Blue	BL	7	Red	RD
2	Orange	OR	8	Black	BK
3	Green	GR	9	Yellow	YL
4	Brown	BR	10	Violet	VI
5	Slate (Grey)	SL	11	Rose (Pink)	RS
6	White	WH	12	Aqua	AQ

Table 6.1: TIA-598-C Color Coding Identification

6.2 Physical Deployment & Site Survey

6.2.1 Site Survey & Planning

Prior to cable deployment, technicians conduct a site survey covering these steps:

- Route Mapping: Finding the shortest, most efficient path from street FAT to house.
- Structural Assessment: Identifying building entry points and optimal indoor ONT placement.
- Safety Check: Keeping clear of electrical mainlines to avoid physical interference.
- BOM Calculation: Generating the exact Bill of Materials (cables, clamps, rosettes).

6.2.2 Outdoor Routing Operations

Aerial Deployment: Fiber cables are strung between utility poles using tension/anchor clamps. Slack loops must be left at ends to accommodate weather-induced physical shifts.

Underground Deployment: In suburban areas, fiber is pulled through buried protective HDPE ducts to shield it from excavation and weather damage.

6.2.3 Indoor Termination

The drop cable passes into the home and terminates inside an Access Terminal Box (ATB) or Fiber Rosette on the wall. A bend-insensitive patch cord (G.657 standard) is then run from the outlet to the ONT, allowing tight bends around walls without causing signal loss.

6.3 Splicing Methods Comparison

Feature	Fusion Splicing	Mechanical / Fast Connectors
Mechanism	Uses a high-voltage electric arc to melt and fuse glass tips permanently.	Physically aligns two fiber ends inside a sleeve using index-matching gel.
Signal Loss	Extremely low (< 0.05 dB to 0.1 dB).	Slightly higher (0.2 dB to 0.5 dB).
Pros & Cons	Most reliable, but requires an expensive fusion splicer machine.	Fast setup, low initial tool cost. Ideal for home drop termination.

Table 6.2: Comparison of Fusion and Mechanical Splicing

6.4 Link Budget & Testing Benchmarks

Single-mode fiber supports bidirectional traffic by using separate wavelengths for upstream and downstream data:

6.4.1 Wavelength Multiplexing

- 1490 nm: Downstream data/voice traffic (OLT to ONT).
- 1310 nm: Upstream data upload traffic (ONT to OLT).
- 1550 nm: Optional downstream video/TV overlay.

6.4.2 Testing Parameters

- Optical Power Meter (OPM): Measures absolute light strength at the subscriber end. Acceptable operational range is between -15 dBm and -25 dBm. Readings below -27 dBm constitute an under-power fault.
- OTDR: Measures backscattered light to map the entire physical link and locate precise coordinates of cuts or defects.

6.5 Troubleshooting & Indicators

Technicians diagnose residential link faults by reading the status LEDs on the ONT:

ONT Indicator	Status State	Meaning / Corrective Action
LOS (Red)	Steady Glow	Loss of Signal: Total physical fiber cut or severe macro-bend. Check outside plant with OTDR.
PON (Green)	Blinking / Flashing	Registering: Authentication in progress. If persistent, light power is weak or configuration profile is missing.
PON (Green)	Steady Solid Glow	Connected: Symmetrical operational link established successfully.
Power Levels	Below -27 dBm	Power Fault: Dirty connector faces or high-loss splice. Rework connection or clean with a OneClick cleaner.

Table 6.3: ONT Troubleshooting Matrix

Chapter 7

Field Visit: BSNL Exchange Office, Manjeri

7.1 Introduction to the Field Visit

To gain practical exposure to telecom systems, we visited the BSNL Telephone Exchange in Manjeri. This field visit provided a real-world look at central office operations. This excursion let us trace the physical flow of data from central carrier routers to the local distribution loops.

7.2 Observations and Core Infrastructure

Technicians at the exchange demonstrated the OLT racks, optical distribution frames (ODF), and multi-fiber termination racks. They explained how incoming voice and data feeds are combined and routed onto fiber cores. The OLT works in tandem with key backend routing and authentication nodes: The BNG (Broadband Network Gateway) handles user authentication, IP address allocation, routing, and internet session management. The MNGN platform provides carrier-grade packet transport, integrating FTTH broadband, IPTV, and business data links onto a unified network. The OLT delivers internet services to ONTs through optical fiber cables.

7.3 Conclusion of the Visit

This technical tour was highly educational, helping us visualize how OLTs and BNGs operate in a real carrier network. It successfully linked our classroom physics to modern telecom infrastructure.

Part Conclusion: Fiber Optic Infrastructure

To summarize, this first section has covered the core architectural principles of modern fiber-optic communications. We studied light propagation in single-mode cores and analyzed fiber structure and attenuation in distribution cables. The rigorous analysis of multi-stage passive optical splitters and mathematical power link budgeting demonstrated how a network ensures operational durability while supporting thousands of subscribers. Practical exercises at Bfone Fibernet allowed us to master fusion splicing, color code identification, and OTDR troubleshooting. Furthermore, the field study at the BSNL Exchange Office in Manjeri provided vital structural insight into macro-scale network topology, illustrating the exact handshake between hardware nodes like OLTs, BNG gateways, and consumer-end ONTs. Ultimately, this module confirmed that efficient FTTH architectures require a precise balance between strict engineering mathematics and rigorous field deployment standards to maintain continuous, high-performance gigabit connectivity.

Part II

Electrical Home Appliances Maintenance and Troubleshooting

Part Introduction: Domestic Electrical Engineering

Electro-mechanical appliances are essential to the comfort and function of modern households. Devices utilizing resistive heating, induction cooking, or single-phase induction motors apply core principles of thermodynamics and electromagnetism. This second module of the report provides a rigorous technical breakdown of consumer appliance electronics, diagnostic logic, and preventive maintenance protocols executed during the industrial training. The goal was to build a solid diagnostic mindset based on electrical theory rather than basic parts-swapping. The chapters ahead map out essential circuit structures (series vs. parallel wiring configurations) and structural protection systems—such as chassis ground earthing, Miniature Circuit Breakers (MCBs), and high-sensitivity Residual Current Circuit Breakers (RCCBs)—designed to mitigate short circuits and hazardous current leakage. Following this safety framework, the module provides an isolated component breakdown, operational analysis, and troubleshooting matrix for seven core appliances: the electric iron, induction cooker, mixer grinder, storage water geyser, ceiling fan, wall fan, and centrifugal water pump. This section concludes with practical calculations on household electricity auditing and charge transport.

Chapter 8

Fundamentals of Electrical Systems & Safety

8.1 Introduction

Household appliances are central to modern life, running constantly to support cooking, ventilation, and water supply. Over time, heavy use causes mechanical wear, insulation breakdowns, and electrical faults. This module was completed to apply electrical theory to practical repair. The course covered safety protocols, wiring layouts, and the step-by-step repair of seven core home appliances. The primary objective was not merely to fix broken devices, but to understand the physics governing their operation—such as electrothermal conversion, electromagnetism, and fluid dynamics—while strictly adhering to industrial safety standards.

8.2 Fundamentals of Electrical Systems & Safety Overview

Prior to starting repairs, we studied basic circuit behaviors and safety components:

8.2.1 Circuit Behaviors & Laws

- Ohm's Law & Power Equation: Diagnostics rely on Ohm's Law ($V = IR$) and the Power formula ($P = VI = I^2R$). These allow us to calculate expected resistance and identify open or short circuits.
- Series Connections: Current remains constant; voltage divides. Used in safety switches, fuses, and thermostats.
- Parallel Connections: Voltage remains constant; current divides. Used in domestic house wiring so that the failure of one appliance does not shut off others.

$$V = IR \quad \text{and} \quad P = VI = I^2R \quad (8.1)$$

8.2.2 Earthing and Protection Systems

- Earthing (Grounding): Connects the metallic chassis of an appliance directly to the earth. In case of insulation failure where a live wire touches the metal body, current diverts safely to the ground instead of passing through a user.

- Fuse / MCB (Miniature Circuit Breaker): Protects the circuit from overcurrent and short circuits by breaking the circuit when current exceeds safe thresholds.
- ELCB / RCCB (Residual Current Circuit Breaker): Detects minute current leakages (as low as 30mA) escaping to ground by comparing hot and neutral currents, tripping instantly if an imbalance is found.

Chapter 9

Technical Breakdown & Diagnostics of Core Home Appliances

9.1 Electric Iron

A. Working Principle

The electric iron operates on Joule's Heating Effect. Current flowing through a high-resistance heating element generates heat proportional to the resistance and square of the current ($H = I^2Rt$). A bimetallic strip thermostat controls the heat. Since the two bonded metals expand at different rates, heating bends the strip to open contacts and cut the circuit. Cooling resets the strip, restarting the heating cycle.

$$H = I^2Rt \quad (9.1)$$

B. Key Components and Parts

- Soleplate: The heavy, polished aluminum or cast-iron base that transfers heat directly to the clothes.
- Heating Element: A ribbon of Nichrome wound around a mica sheet insulated to prevent electric shock while maximizing heat transfer.
- Bimetallic Thermostat: The temperature control mechanism equipped with an adjustable knob to set specific heat thresholds.
- Thermal Fuse: A backup safety device that permanently melts and opens the circuit if the thermostat fails.

C. Appliance Conclusion

Studying the iron showed the practical side of Joule heating and thermal feedback loops. Working on it highlighted the role of thermal fuses in protecting against fire hazards.

9.2 Induction Cooker

A. Working Principle

Unlike standard stoves that use heat conduction, induction cooktops operate using electromagnetic induction and eddy currents. An internal inverter converts 50Hz AC mains into

high-frequency AC (20-40kHz). This current flows through an induction coil to generate an alternating magnetic field. Placing a ferromagnetic pan on the glass induces eddy currents in the pan's base. The pan's electrical resistance causes it to heat up directly, leaving the glass top relatively cool. **B. Key Components and Parts**

- Induction Coil: A flat, spirally wound copper coil that generates the alternating magnetic field.
- IGBT (Insulated Gate Bipolar Transistor): The high-speed electronic switch that modulates power and creates the high-frequency current.
- Main PCB & Microcontroller: The 'brain' that monitors user inputs, manages power delivery, and reads sensor data.
- Cooling Fan & RTD (Resistance Temperature Detector): Critical elements for heat dissipation and monitoring pan temperatures to prevent dry boiling.

C. Appliance Conclusion

Induction stove troubleshooting offered great experience with solid-state inverters and magnetic heating. It showed the importance of tracing faults at the board component level.

9.3 Mixer Grinder

A. Working Principle

Mixer grinders use an AC Universal Motor to convert electrical energy into rotation. The universal motor is series-wound and runs on either AC or DC. Current flows through the stator field coils and the rotor windings through carbon brushes. The resulting magnetic fields produce high starting torque and speeds up to 20,000 RPM, ideal for grinding and blending. **B. Key Components and Parts**

- Stator & Rotor (Armature): The fixed and rotating electromagnetic components that generate mechanical torque.
- Commutator & Carbon Brushes: The assembly that transfers electrical current from the stationary wires to the rotating armature.

9.3.1 Ceiling Fan

A. Working Principle

Ceiling fans utilize a Single-Phase Permanent Split Capacitor Induction Motor. Since single-phase AC cannot self-start a motor, the stator features main (running) and auxiliary (starting) windings. A start/run capacitor in series with the starting winding shifts the current phase by 90°. This phase difference creates a rotating magnetic field, inducing current in the rotor and turning the fan blades. **B. Key Components and Parts**

- **Stator & Rotor:** Stationary inner core with starting/running windings and the rotating outer casing.
- **Run/Start Capacitor:** Typically 2.25 μ F to 2.5 μ F capacitor used to split the phase and start rotation.
- **Ball Bearings:** Precision bearings that reduce mechanical friction, ensuring smooth and quiet operation.
- **Down-Rod Assembly:** Structural hardware used to safely suspend the fan from the ceiling.

C. Appliance Conclusion

Repairing ceiling fans helped us understand phase-splitting in single-phase motors. The troubleshooting process showed how to distinguish winding issues from mechanical bearing wear.

9.3.2 Wall Fan

A. Working Principle

Wall fans also use a single-phase capacitor induction motor but add mechanics for oscillation. The main motor drives the blades, while a mechanical gear linkage or a small secondary motor rotates the fan assembly side-to-side. Fan speed is controlled by switching between motor winding taps or using a TRIAC circuit to vary voltage.

B. Key Components and Parts

- **Fan Blades and Guard:** Aerodynamically pitched plastic or metal blades protected by a wire mesh safety cage.
- **Oscillation Gear Assembly / Clutch Knob:** A mechanical gear train connected to the rotor shaft that translates high-speed rotation into slow, wide-angle swiveling motion.
- **Speed Control Switch (Pull-Cord / Push-Button / Remote PCB):** Mechanisms to toggle between different motor speed taps or control electronic voltage regulation.
- **Thermal Cut-Off (TCO):** A compact thermal fuse embedded directly inside the motor windings to protect the motor from burning out if the blades get obstructed or jammed.

C. Appliance Conclusion

Working on the wall fan showed how mechanical gearboxes connect to electrical motors. It was a good lesson in motor winding safety fuses and fixing mechanical oscillation failures.

Chapter 10

Troubleshooting & Maintenance Practice Matrix

ONT Indicator	Status State	Meaning / Corrective Action
LOS (Red)	Steady Glow	Loss of Signal: Total physical fiber cut or severe macro-bend. Check outside plant with OTDR.
PON (Green)	Blinking / Flashing	Registering: Authentication in progress. If persistent, light power is weak or configuration profile is missing.
PON (Green)	Steady Solid Glow	Connected: Symmetrical operational link established successfully.
Power Levels	Below -27 dBm	Power Fault: Dirty connector faces or high-loss splice. Rework connection or clean with a OneClick cleaner.

Table 10.1: ONT Troubleshooting Matrix Summary

Appliance	Common Symptom	Root Cause	Detected	Corrective Action Performed
Electric Iron	No heating; indicator light off	Open circuit in thermal fuse / Broken power cord		Replaced thermal fuse; trimmed and re-terminated power cord.
Induction Cooker	Dead display; won't power up	Blown main fuse due to shorted IGBT		Replaced the shorted IGBT and main glass fuse; checked gate driver circuit.
Mixer Grinder	Sparks from bottom; sudden stop	Worn carbon brushes / Over-load switch tripped		Replaced carbon brushes; cleared jar jam and reset the bottom OLP switch.
Water Heater	Water not heating; indicator on	Scale buildup or open-circuit in heating element		Drained tank, removed limescale, and replaced the immersion heating element.
Ceiling Fan	Humming sound; spins very slowly	Degraded or leaked capacitor		Desoldered/disconnected old capacitor, verified value, and installed new $2.5\mu\text{F}$ unit.
Wall Fan	Fan rotates but does not oscillate / Over-heating	Stripped linkage gears / Dry rotor shaft		Replaced the plastic gear wheel assembly; cleaned and lubricated the rotor shaft with machine oil.
Pump Motor	Motor hums but does not rotate	Seized bearings / Defective starter capacitor		Freed impeller manually; replaced bearings and capacitor.

Table 10.2: Appliance Troubleshooting & Maintenance Practice Matrix

10.1 Internship Outcomes & Skills Acquired

- **Safety Instinct Diagnostics:** Developed a strict protocol-first mindset—always verifying isolation with a neon tester, utilizing insulation mats, and ensuring appropriate grounding before applying mains voltage.
- **Systematic Fault Profiling:** Shifted away from random guesswork to structured trou-

troubleshooting: tracing faults from the source point (plug/cord) through protection systems (OLPs/fuses) to control systems and finally the load.

- Instrument Competency: Attained fluid precision in utilizing digital multimeters to measure resistance drops, AC loop voltages, and verifying open/short circuits.

Chapter 11

Applied Practical Calculations

11.1 Domestic Electricity Consumption Auditing

Domestic electricity usage is measured in kilowatt-hours (kWh), with one unit of electricity equal to 1 kWh.

Calculate the 30-day energy consumption for a home operating two 40W tubes for 6 hours/day, three 65W fans for 8 hours/day, and auxiliary loads of 150W for 2 hours/day.

$$E_{\text{Lights}} = 2 \times 40 \text{ W} \times 6 \text{ hrs/day} \times 30 \text{ days} = 14,400 \text{ Wh} = 14.4 \text{ kWh}$$

$$E_{\text{Fans}} = 3 \times 65 \text{ W} \times 8 \text{ hrs/day} \times 30 \text{ days} = 46,800 \text{ Wh} = 46.8 \text{ kWh}$$

$$E_{\text{Misc}} = 150 \text{ W} \times 2 \text{ hrs/day} \times 30 \text{ days} = 9,000 \text{ Wh} = 9.0 \text{ kWh}$$

$$E_{\text{Total}} = 14.4 \text{ kWh} + 46.8 \text{ kWh} + 9.0 \text{ kWh} = 70.2 \text{ kWh (Units)}$$

11.2 Charge Calculation in Audio Systems

An appliance connected to a 230V line consumes 8 Megajoules (MJ) of energy. Calculate the total electric charge that passed through the system.

$$\text{Given: } V = 230 \text{ V, } E = 8 \times 10^6 \text{ J}$$

$$\text{Equation: } E = qV$$

$$\begin{aligned} q &= \frac{E}{V} \\ &= \frac{8 \times 10^6 \text{ J}}{230 \text{ V}} \\ &= 3.48 \times 10^4 \text{ Coulombs (C)} \end{aligned}$$

Part Conclusion: Domestic Electrical Engineering

In summary, this section connected classroom physics with real-world electrical troubleshooting. Studying the components and wiring of seven core appliances provided valuable insight into domestic electrical layouts. Tracing diagnostic signatures—such as a bimetallic contact opening in a thermal loop or a split-phase capacitor drop in an induction motor—showed how equipment behaviors can be accurately tracked using standard multimeter analysis. The exercises emphasized the importance of safety devices like thermal cut-offs (TCOs) and overload protectors (OLPs) in preventing damage. Additionally, the quantitative practical math problems confirmed that accurate load auditing and tracking energy changes are key to managing household electricity efficiently. Ultimately, the mechanical repair skills and systemic troubleshooting methods gained during this maintenance module form a critical practical toolkit, ensuring that home environments remain completely safe and operationally productive.